Trust Flow-Driven Trusted Federated Learning in Edge Computing: A Study on Key Technologies

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Abstract

In recent years, federated learning has become a key enabler of edge intelligence by supporting collaborative model optimization under privacy constraints through the paradigm of keeping data local and exchanging only model parameters. However, in open edge-computing environments, the combined effects of non-independent and identically distributed (Non-IID) data, device heterogeneity, and untrusted clients expose significant tradeoffs among security, robustness, and generalization for existing robust aggregation methods. To address this issue, this paper proposes a trust-flow-driven trusted federated aggregation mechanism that covers client admission, runtime monitoring, and secure aggregation from a hardware-software co-design perspective. First, remote attestation and runtime monitoring based on trusted execution environments (TEEs) are integrated to enforce client admission and verify execution integrity during local training. Second, to identify stealthy attacks across training rounds, this paper designs an orthogonal dual-stream reputation evolution method centered on client trust states, decomposing them into a historical utility stream for long-term contribution and an instant risk stream for short-term anomalous behavior, thereby providing persistent dynamic trust signals for subsequent auditing and aggregation control. Third, this paper proposes a layer-wise differentiated aggregation strategy guided by a clean reference direction. By combining layer-wise gating, weight re-normalization, and dynamic clipping, the method suppresses malicious updates while preserving legitimate heterogeneous contributions, thereby mitigating the degradation of generalization under Non-IID settings. Experiments on Flower with CIFAR-10 under 20 clients and six mixed attack types show that the proposed mechanism achieves an average accuracy of 92.31% (± 0.09%) while limiting the attack success rate (ASR) to 10.21% (± 0.05%), with the permanent false positive rate on benign clients approaching zero.

Keywords: Federated Learning; Edge Computing; Trusted Execution Environment; Risk State Modeling; Layered Robust Aggregation; Backdoor Defense

1 Introduction

Current advancements in terminal device computing power enable efficient local model training using on-device data. Meanwhile, to better aggregate multi-device training experiences for collaborative model optimization while preserving local data privacy, the federated learning paradigm has been extensively studied [1]. Its core feature—exchanging only model parameters without offloading data—is widely applied in distributed collaborative training, serving as a critical pillar for edge intelligence. However, as the interconnected network environment becomes increasingly open, the performance bottlenecks of federated learning no longer reside solely in model architecture or optimizer design. The trustworthiness and robustness of the aggregation phase have emerged as vital factors determining overall availability. In practical scenarios, edge devices often exhibit significant heterogeneity in both data distribution and processing capabilities. Furthermore, fully constraining the behaviors of participating clients remains challenging. Such characteristics inevitably amplify statistical drifts and introduce security risks during empirical aggregation.

In standard federated learning paradigms, clients upload parameters after completing local data processing and model training. The server then aggregates these local updates and distributes the updated global model, iterating this process until convergence. Evidently, this workflow encounters multiple vulnerabilities within the increasingly open, multi-edge interconnected network. Open training environments inherently lack mandatory integrity constraints. Malicious clients can bypass actual training, directly crafting and uploading poisoned parameters, thereby severely impairing the server's ability to verify the authenticity of aggregated updates. Several studies propose methods based on trusted execution or verifiable training [2, 3, 4], employing hardware-assisted remote attestation to ensure local training authenticity. Nevertheless, these mechanisms fail to address scenarios where the training process is genuine but the model updates are injected with malicious features. Furthermore, attackers execute parameter poisoning or stealthy backdoor attacks [5, 6] by embedding malicious triggers within normal update distributions, drastically reducing the success rate of single-round detection. To alleviate the instability of single-round assessments, researchers have introduced methods based on historical reputation and similarity [7, 8] that leverage cross-round information. Yet, these approaches suffer from delayed recognition of sleeper attacks, where adversaries accumulate reputation early on before engaging in persistent, low-amplitude poisoning. Additionally, single-reputation scores combined with fixed-threshold mechanisms remain highly inadequate for identifying deep abnormal trigger features.

The presence of non-independent and identically distributed (Non-IID) data across devices increases the directional dispersion of optimal updates. This heterogeneity easily blurs the boundary between normal statistical fluctuations and anomalous deviations, creating a conflict between robustness and generalization under strong Non-IID conditions. Certain approaches rely on distance-based exclusion or statistical truncation to suppress abnormal updates [9, 10], while others adaptively adjust weights to mitigate heterogeneous impacts [11]. Under severe long-tail distributions, however, such methods often misclassify legitimately drifted parameter updates as anomalies. This misjudgment can lead to the excessive exclusion of crucial heterogeneous information, significantly degrading the model's generalization capability. In real-world edge intelligence scenarios, these challenges—open execution environments, stealthy cross-round attacks, and pronounced heterogeneous statistical drifts—rarely occur in isolation. Multiple factors frequently compound, leading to intractable dilemmas such as elevated false positive rates from tightening interception thresholds or increased false negative rates when relaxing them. Consequently, establishing a unified state modeling and coordinated decision-making mechanism spanning the entire workflow remains a critical challenge. Such a mechanism must balance client admission and trustworthiness during local training with low false positive and high recall requirements during aggregation, all while maintaining the convergence and generalization performance of collaborative model optimization.

To tackle the three major challenges outlined above, this paper proposes a trusted federated aggregation mechanism driven by a full-process trust flow, conceptualized from a hardware-software co-design perspective. This mechanism follows the execution pipeline of federated learning. During client admission and local training, TEE remote attestation, runtime feature entropy analysis, and Kalman filtering are introduced to formulate a dynamic admission trust score that updates across rounds. At the server auditing stage, the current-round content quality is evaluated based on a pure reference direction. The client's state is then decoupled into a historical utility flow and an instant risk flow, utilizing Bayesian posterior inference and risk EMA to separately characterize long-term contributions and short-term anomalies. During the aggregation phase, layered risk gating, survivor weight re-normalization, and dynamic clipping are executed based on these trust states. This strategy suppresses malicious updates while retaining legitimate heterogeneous contributions, effectively mitigating the tradeoff between security and generalization under Non-IID conditions.

The primary contributions of this paper are:

1. **Proposing a TEE-anchored dynamic admission mechanism for edge clients.** Remote attestation defines the client admission boundary. Variations in gradients, losses, and instruction distributions during training are then transformed into entropy features. By leveraging Kalman filtering to update the $TrustScore$, admission control extends from a one-time static verification to a continuously evolving trust metric.

2. **Developing a dual-stream orthogonal reputation evolution mechanism against stealthy attacks.** Client states are decoupled into a historical utility stream (HistPerf) and an instant risk stream (RiskEMA), independently handling long-term contributions and sudden anomalies. A $RawScore$ connects content auditing, state transitions, and subsequent

aggregation control, balancing malicious recall with the suppression of permanent false positives.

3. **Designing a layered risk gating aggregation strategy tailored for strong Non-IID scenarios.** Guided by a server-side clean reference direction, layer-wise security sensitivities, and dynamic L2 clipping, this approach applies differentiated admission thresholds and survivor weight re-normalization across network layers. It reduces the risk of deep backdoor penetration while preserving valid updates from legitimate long-tail nodes.

4. **Implementing and validating the framework end-to-end under mixed attack scenarios.** Experiments involving 20 clients and 6 mixed attack types ($n = 5$) were conducted on the Flower platform. Results demonstrate that under high-adversarial settings, the proposed framework maintains an accuracy of 92.31%(±0.09%) and restricts the ASR to 10.21%(±0.05%), with the permanent false positive rate approaching zero.

2 Background and Related Work

2.1 Federated Learning and Edge Computing Collaborative Architecture

Federated learning (FL) completes global model training through parameter-level collaboration under the constraint that data remain local. In edge computing scenarios, the client-edge-cloud architecture has been widely adopted: the edge side performs local training and uploads updates, while the central server handles aggregation and model distribution.

In the standard FL setting, the global optimization objective is to minimize the empirical risk over the data distributions of all clients:

$$\min_{W} \mathcal{L}(W) = \sum_{k=1}^{K} p_k \mathcal{L}_k(W) \tag{1}$$

where $p_k = |\mathcal{D}_k| / \sum |\mathcal{D}_j|$ denotes the data-volume proportion. In round t, the server distributes the global model $W_{global}^{(t-1)}$. Client k takes it as the initial model and runs SGD for E local epochs with learning rate η_l, yielding the local update:

$$\Delta W_k^{(t)} = W_{k,E}^{(t)} - W_{global}^{(t-1)}, \quad \text{where } W_{k,e}^{(t)} = W_{k,e-1}^{(t)} - \eta_l \nabla \mathbb{E}[\ell(W_{k,e-1}^{(t)}; x, y)] \tag{2}$$

Here, the expectation $\mathbb{E}[\cdot]$ denotes mini-batch sampling over the local dataset $\mathcal{D}_k$. The client then uploads its update, and the server applies a form of parameter aggregation: $W_{global}^{(t)} = W_{global}^{(t-1)} + \text{Agg}(\{\Delta W_k^{(t)}\}_{k\in\Phi})$.

This architecture, however, naturally faces communication latency, device heterogeneity, and Non-IID data. To mitigate these issues, prior work has proposed methods such as Clustered FL [12], adaptive pruning-expansion (FedPE) [13], and split federated learning (ParallelSFL) [14]. As system complexity increases, the attack surface also expands, giving malicious nodes more room to remain hidden and interfere with training.

2.2 Edge Impersonation, Model Poisoning, and Stealthy Backdoor Attacks

In open federated networks without mandatory admission control, the server has difficulty accurately judging client identities and update quality. Attacks mainly fall into two categories. The first is **untargeted poisoning**, such as Sign-flipping and Gradient Scaling, which aims to disrupt global convergence and cause denial of service. The second is **targeted backdoor** attack, including label flipping, clean-label poisoning, and semantic backdoors, which induce targeted misclassification at inference time through concealed triggers.

2.3 Mechanisms and Inherent Limitations of Existing Active Defenses

To address these threats, early defenses largely relied on geometric and statistical filtering, such as Krum[9] and Trimmed Mean[10]. More recent studies have introduced temporal trust evaluation, including FLTrust[7], FoolsGold[8], and RaSA[11], while continuing to expand into malicious detection[15], trusted training frameworks[2], hierarchical protection[16], and related directions. For backdoor and collusion attacks, schemes such as FLPurifier[17], RoseAgg[18], and ShieldFL[19] have also been developed. Purely software-based defenses still have clear weaknesses under strong Non-IID conditions and high-privilege attacks. Tightening thresholds raises the false positive rate (FPR), while relaxing them may allow dormant attacks to pass. When attackers control the underlying environment, software probes that lack a hardware root of trust also struggle to form a complete defensive loop.

2.4 Trusted Hardware Root of Trust in Federated Environments (TEE)

To fill the gap in physical trust, research has begun to introduce trusted execution environments (TEEs) such as Intel SGX and ARM TrustZone. Through isolated execution and protected memory, TEEs preserve the integrity of critical code and key material, making it difficult for an attacker to directly tamper with the attestation process even after the host operating system has been compromised[3, 20]. Existing work shows that TEEs have practical value in training integrity[4], threat mitigation[21], and trust sharing in industrial IoT[22]. Their federated extensions and efficiency optimization in SGX environments have also been validated[23, 24]. On this basis, this paper builds a hardware-software collaborative defense system. Unlike methods that only perform static encryption or single-dimensional anomaly evaluation, this work temporally couples hardware remote attestation with dual-stream reputation evolution. The cloud side is responsible for parameter auditing, while the edge-side TMAA provides trusted evidence and runtime monitoring, forming a coordinated protection mechanism that combines mandatory admission with dynamic circuit breaking.

3 System Model and Problem Definition

3.1 Client-Edge-Cloud Collaborative Architecture and System Workflow Modeling

This paper considers a typical client-edge-cloud heterogeneous communication environment (see Figure 1). The system consists of a central **aggregation server (Server, $\mathcal{S}$)** and K edge clients, denoted by $\mathcal{C} = \{c_1, c_2, \ldots, c_K\}$. Clients may join or leave dynamically, and some nodes may be controlled by attackers.

Based on the network interaction topology above, the proposed trust-flow-driven trusted federated framework builds a complete trust transmission and evaluation chain between the edge and the cloud. The overall workflow of the system and the relationships among its main physical entity modules are defined as follows:

1. **Edge client side (trusted execution and feature extraction)**: Each admitted client must be equipped with a trusted execution environment (TEE) and a trusted management agent (TMAA) deployed inside it. During local training, the TMAA generates an Attestation Quote before training to statically verify the integrity of the execution environment. It also continuously monitors key behavioral feature sequences during training, such as changes in gradient direction and loss fluctuation. Information entropy and Kalman filtering are then used for an initial dynamic trust assessment, which is uploaded to the cloud together with the model update parameters.

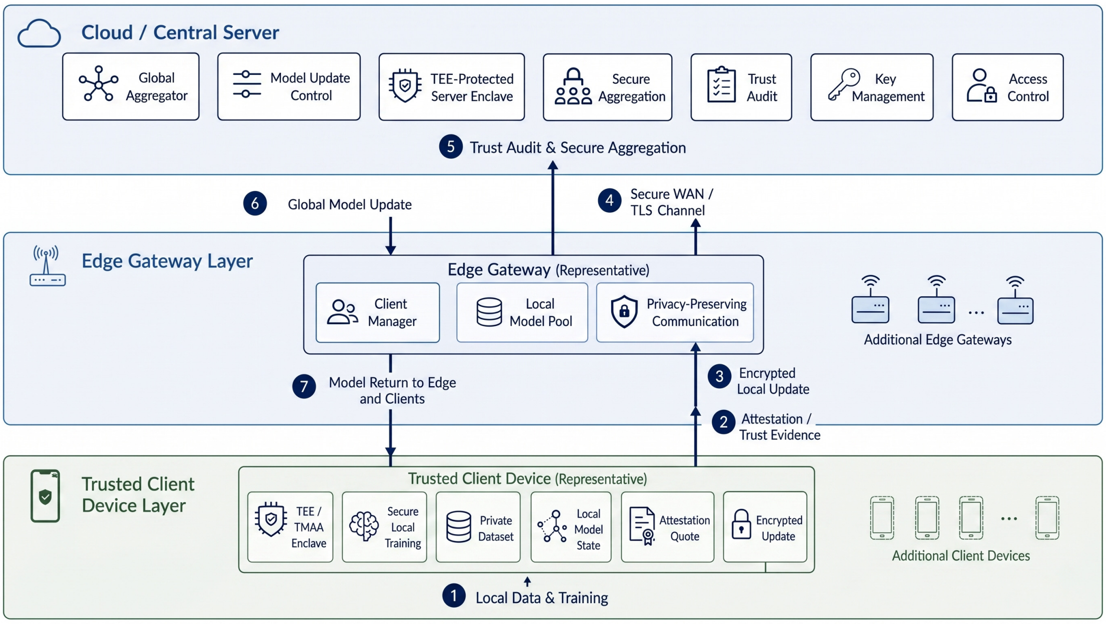

Figure 1: Federated computing network interaction: client-edge-cloud collaborative physical architecture based on TEE-anchored encryption

2. **Central server side (auditing, evolution, and aggregation control)**: As the global coordinator, the server receives client updates and trust evidence, then performs a three-stage coordinated defense based on trust flow. It first conducts **content auditing**, extracting a clean reference direction and using cosine similarity to calculate high-dimensional update quality. It then enters **dual-stream state evolution**, where Beta Bayesian inference is used to build a historical utility stream for long-term contribution evaluation, while multidimensional max-value probes update an instant risk stream to identify sudden anomalies. Finally, **layered robust aggregation** applies differentiated gated clipping and survivor weight renormalization according to the security sensitivity of different network layers, producing and distributing a globally secure update.

Through the coordinated interaction of the edge and cloud entity modules, the system forms a full-lifecycle physical and logical closed loop: hardware admission control $\rightarrow$ runtime feature monitoring $\rightarrow$ trust state evolution $\rightarrow$ risk-gated aggregation.

3.2 Composite Threat Assumptions and Strong Trust-Base Boundary (Threat Model)

To clearly define the boundary between attacker and defender capabilities, this paper adopts the following threat model (see Figure 2):

- **Hardware isolation boundary (root of trust)**: Under standard security assumptions, excluding advanced attacks such as microarchitectural side channels and physical probing[3], TEEs such as TrustZone/SGX can preserve the integrity of critical execution and attestation processes even if the host operating system is compromised.
- **Honest-but-curious server**: The server follows the protocol for dual-stream scoring and layered aggregation, but it may attempt to infer additional private information about clients.
- **Internal malicious clients (Internal Attackers/Poisoners)**: Attackers may possess valid admission identities and locally perform label tampering, sample poisoning, sign flipping, gradient scaling, and related operations. This paper assumes an upper bound of $< 50\%$

malicious clients. The main experiments use 30% as a typical high-risk setting and additionally report a 50% boundary stress test.

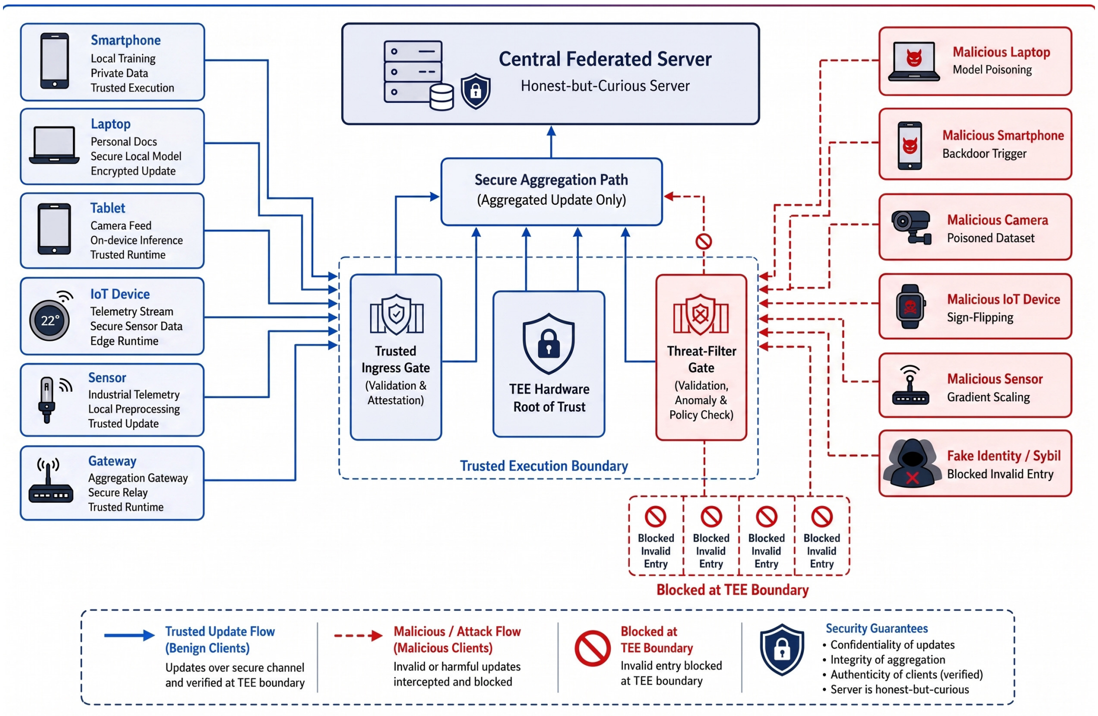

Figure 2: Security penetration and multidimensional composite poisoning threat model in an edge wide-area network

3.3 Core Defense Optimization Objectives of the Security Architecture (Design Goals)

For highly heterogeneous data and a high proportion of malicious participants, the design goals of the proposed framework are as follows:

1. **High defense effectiveness and robust convergence**: Under multiple types of poisoning and backdoor attacks, the system must accurately identify malicious nodes, namely achieving high TPR and low ASR, while maintaining high availability and stable convergence of the global main-task model.

2. **High tolerance for heterogeneous long-tail nodes**: The framework should effectively overcome feature deviation caused by Non-IID data distributions, avoid mistakenly removing honest nodes that carry long-tail data, and keep the false positive rate (FPR) close to zero.

3. **Lightweight overhead and edge adaptability**: Without introducing the computational burden of full-scale cryptography, such as fully homomorphic encryption, the framework should build an efficient trusted protection mechanism whose computation and communication overheads remain compatible with the resource bottlenecks of heterogeneous edge devices.

4 Proposed Method: Trust-Flow Framework

For the threat model above, this paper proposes a trust-flow-driven trusted federated learning framework (see Figure 3). Following the execution order of federated learning, the framework first performs trusted admission and runtime monitoring on the client side, then conducts current-round content auditing and cross-round reputation evolution on the server side, and finally enters layered security control during aggregation. Each training round is therefore divided into five phases: pre-training admission, content auditing, state evolution, layered aggregation, and global distribution. At a higher level of abstraction, these phases correspond to three layers: client admission and monitoring, server auditing and evolution, and aggregation security assurance.

The key of this macro-to-micro structure is the continuous transfer of trust states. Phase one outputs $TrustScore$, indicating whether the client has a trustworthy basis for admission and execution. Phase two outputs $ContentScore$, judging whether the update submitted in the current round deviates from the clean reference and the collaborative direction of the group. Phase three deposits current-round signals into $HistPerf$ and $RiskEMA$, distinguishing long-term low contribution from short-term high risk. Phase four fuses the three types of signals into $RawScore$ and performs differentiated aggregation by network layer. Unlike single-point defenses, data generated at each phase continues to constrain later phases, continuously updating the **Trust Flow** and forming closed-loop control.

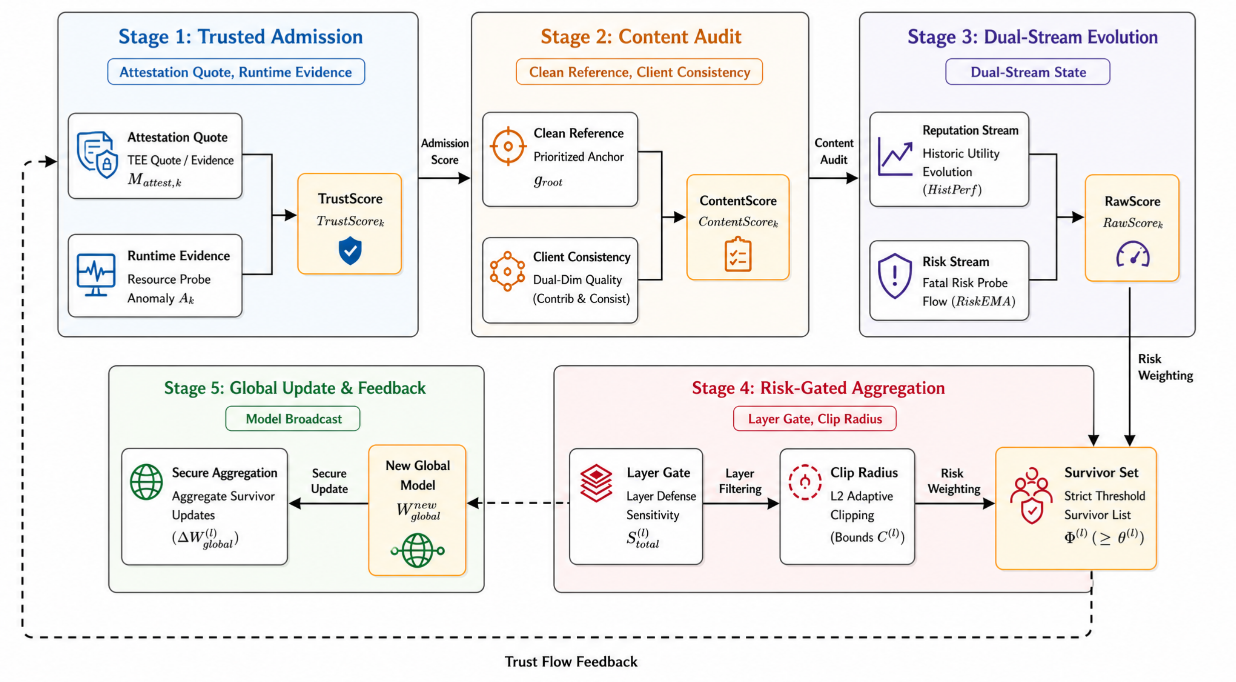

Figure 3: Overview of the five-phase defense closed loop in trust-flow-driven trusted federated learning

To facilitate the following derivation, Table 1 lists the definitions of core symbols.

4.1 Hardware Admission and Dynamic Trust Assessment

Client admission should not be limited to one-time identity verification. It requires a tracking mechanism that reflects the continuing trustworthiness of local training. For this purpose, the initial trust level is decoupled into a two-layer architecture of static admission and dynamic evaluation. The former relies on underlying hardware to judge the legitimacy of the execution environment, while the latter draws on the robust filtering idea behind the maximum correntropy Kalman filter[25] to convert transient behavioral fluctuations during training into dynamic observation signals, producing a transferable $TrustScore$ through filtering.

Table 1: Core system parameters and trust-flow notation

Symbol	Definition	Symbol	Definition
$W_{global}^{(t)}$	Global model weights in round t	$\mathcal{A}$	Active node set eligible for valid aggregation in the current round
ΔW_k	Model gradient/parameter update submitted by client k	$\Phi^{(l)}$	Valid secure survivor subset at layer l
$TrustScore_k$	TEE hardware-anchored admission trust score	$HistPerf_k$	Accumulated state of the historical utility stream ($HistPerf \in [0, 1]$)
$ContentScore_k$	Current-round content quality score under dual references	$RiskEMA_k$	Decayed state of the instant risk stream ($RiskEMA \in [0, 1]$)
$RawScore_k$	Final privilege score after fusing three-dimensional scores and applying risk discounting	$M_{attest,k}$	Hardware remote attestation verification flag
g_{root}	Clean guiding anchor direction extracted by the server	A_k	Runtime anomaly score of client k in physical resource usage

1. **Static Hardware Admission**: During the initialization of federated computing, each participating node must invoke its internally isolated device-unique key (DUK) and initiate a remote attestation quote to the central server, including the chain values of platform configuration registers (PCRs). The server verifies memory code segments and runtime signatures, then grants initial communication privileges to nodes that pass verification ($M_{attest,k} = 1$).

2. **Feature Sequence Extraction and Entropy Analysis**: Related multidimensional trust-evaluation work in mobile crowdsensing shows that a single static trust measure is insufficient to capture context-dependent reliability[26]. Inspired by this idea, the client-side TEE is configured to continuously monitor key feature sequences during the training cycle, $\mathcal{X}_k = \{\Delta\text{grad}, \Delta\text{loss}, \text{instr\_dist}\}$. The information entropy $H(\mathcal{X}_k)$ of this sequence is quantified as:

$$H(\mathcal{X}_k) = -\sum p(x_i) \log p(x_i) \tag{3}$$

   A high entropy value in the sequence usually indicates greater uncertainty in system operation. In physical terms, this often corresponds to backdoor-controlled execution, severe disturbance in gradient direction, or abnormal underlying instruction distributions.

3. **ITL-Driven Dynamic Trust Evolution (Kalman-based)**: To prevent abrupt anomalies from contaminating the trust state, this framework constructs an entropy-driven adaptive covariance mapping mechanism. Instead of assuming a constant prior, the system maps the dynamically captured information entropy to the measurement-noise covariance matrix $R_k = f(H(\mathcal{X}_k))$ in the Kalman observation equation. In the classical predict-update recursion:

   - **Prediction stage**: Based on the posterior distribution from the previous round, the prior trust expectation $\hat{T}_k^-$ at the current moment is extrapolated.

- **Update stage**: The instantaneous behavioral features of the current round are introduced for measurement update, yielding the posterior trust estimate $\hat{T}_k$ and the corresponding state error covariance matrix P_k.

On this basis, combined with the high-entropy penalty property of anomalous sequences, the system introduces the following core state evolution rule:

$$\hat{T}_k^{(t)} = \hat{T}_k^{(t-1)} + K_k^{(t)}(Z_k^{(t)} - \hat{T}_k^{(t-1)}) \tag{4}$$

$$R_k^{(t)} = \exp(\lambda \cdot H(\mathcal{X}_k^{(t)})) \tag{5}$$

In these equations, $Z_k^{(t)}$ maps the node's real-time baseline observed utility. When client behavior undergoes a high-risk jump, namely high entropy, the exponentially amplified $R_k^{(t)}$ rapidly reduces the Kalman gain $K_k^{(t)}$. This mechanism makes the filter automatically suppress the absorption of high-risk anomalous instantaneous observations, thereby maintaining the stable resilience of the posterior trust estimate.

Furthermore, by the mathematical nature of Bayesian filtering, the covariance matrix P_k iteratively produced by the filter objectively characterizes the confidence boundary of the node's dynamic trust estimate. A smaller P_k indicates that the current filter has more effectively suppressed measurement noise and that the reputation score is more reliable. Phase one therefore does not output a coarse binary decision. Instead, it jointly evolves the expected mean and uncertainty produced by Kalman filtering, forming a temporally resilient admission privilege factor $TrustScore_k = \hat{T}_k^{(t)} \cdot (1/(1 + P_k^{(t)}))$.

4.2 Reference Direction Construction and Consistency Auditing

Relying only on cosine similarity among clients can be misled by collusion attacks. To address this issue, this paper introduces **Vanilla Reference Prioritization**. When the server holds a small isolated validation set (Pure Clean Set), it first computes g_{root_clean} as the reference direction. If this direction is unavailable, a fallback reference g_{ref} is constructed from updates submitted by high-trust, low-risk clients. The corresponding weight is defined as:

$$\omega_k^{ref} = TrustScore_k \cdot (1 - RiskEMA_k^{prev})^2. \tag{6}$$

The final reference direction g_{root} is:

$$g_{root} = \begin{cases} g_{root\_clean}, & \text{if server validation direction available,} \\ \sum_k \frac{\omega_k^{ref}}{\sum_j \omega_j^{ref}} \Delta W_k, & \text{otherwise.} \end{cases} \tag{7}$$

After obtaining g_{root}, the server calculates a content quality score for each client update ΔW_k. This score considers both global contribution, namely consistency with the reference direction, and group consistency, namely the degree of agreement with other high-reputation clients:

$$S_{contrib,k} = \max\left(0,\ \alpha_1 + \alpha_2 \cdot \cos(\Delta W_k, g_{root})\right), \tag{8}$$

$$S_{consist,k} = \frac{\sum_{j \neq k} TrustScore_j \cdot \max\left(0,\ \alpha_1 + \alpha_2 \cdot \cos(\Delta W_k, \Delta W_j)\right)}{\sum_{j \neq k} TrustScore_j + \varepsilon}, \tag{9}$$

$$ContentScore_k = \frac{(1 + \beta_{fusion}^2) \cdot S_{consist,k} \cdot S_{contrib,k}}{\beta_{fusion}^2 \cdot S_{consist,k} + S_{contrib,k} + \varepsilon}. \tag{10}$$

Here, $\alpha_1 + \alpha_2 = 1.0$ adjusts the cosine mapping interval. The parameter $\beta_{fusion} > 1.0$, with default $\beta_{fusion} = 2.0$, moderately increases the weight assigned to consistency with the clean direction, strengthening resistance to collusive drift. $ContentScore_k$ describes only the content quality of the current-round update and is not equivalent to permanent reputation. For legitimate long-tail nodes under strong Non-IID settings, a low single-round score may result from data skew rather than attack behavior. This score is therefore passed to the cross-round state evolution in phase three, where subsequent handling is jointly determined by the historical utility stream and the instant risk stream.

Algorithm 1 Admission and content auditing based on TEE hardware anchors and clean reference

Input: $\mathcal{C}$, $\{\Delta W_k\}_{k\in\mathcal{C}}$, $RiskEMA^{(t-1)}$
Output: $TrustScore_k$, $ContentScore_k$, valid reference baseline g_{root}

Phase 1: Hardware Anchoring and Dynamic Trust Measurement
for each client $k \in \mathcal{C}$ **do**
  if $M_{attest,k} == 0$ (TEE remote attestation fails) **then**
    Reject admission: $TrustScore_k \leftarrow 0$, and skip this round
  else
    Inside TEE: record behavior sequence $\mathcal{X}_k = \{\Delta\text{grad}, \Delta\text{loss}, \text{instr\_dist}\}$
    Compute information entropy $H_k \leftarrow -\sum p(x_i)\log p(x_i)$
    Map entropy to measurement noise $R_k \leftarrow \exp(\lambda \cdot H_k)$
    Kalman filter update:
    Trust estimate $\hat{T}_k$, covariance $P_k \leftarrow \text{KalmanFilter}(H_k, \text{previous state})$
    Admission trust score $TrustScore_k \leftarrow \hat{T}_k \cdot \frac{1}{1+P_k}$
  end if
end for
Phase 2: Reference Direction Construction and Collaborative Consistency Auditing
if server clean validation data are available **then**
  Extract true guiding gradient $g_{root} \leftarrow g_{root\_clean}$
else
  Compute nonlinear weight $\omega_k^{ref} \leftarrow TrustScore_k \cdot (1 - RiskEMA_k^{(t-1)})^2$
  Generate fallback clean anchor $g_{root} \leftarrow \sum_k(\omega_k^{ref}\Delta W_k)/\sum_j \omega_j^{ref}$
end if
for each admitted active client k **do**
  Global contribution score $S_{contrib,k} \leftarrow \max(0, \alpha_1 + \alpha_2\cos(\Delta W_k, g_{root}))$
  Group collaboration score $S_{consist,k} \leftarrow \frac{\sum_{j\neq k} TrustScore_j \max(0,\alpha_1+\alpha_2\cos(\Delta W_k,\Delta W_j))}{\sum_{j\neq k} TrustScore_j+\varepsilon}$
  Harmonic content score $ContentScore_k \leftarrow \frac{(1+\beta_{fusion}^2)S_{consist,k}S_{contrib,k}}{\beta_{fusion}^2 S_{consist,k}+S_{contrib,k}+\varepsilon}$
end for
return $\{TrustScore_k, ContentScore_k, g_{root}\}$

4.3 Dual-Stream Evolution of Historical Utility and Instant Risk

In edge computing scenarios characterized by strong Non-IID (non-independent and identically distributed) data, the conventional "low-score equals elimination" strategy frequently conflates "low utility induced by data long-tailness" with "high risk caused by malicious poisoning," leading to severe false positive rates (FPR). To disentangle this dilemma, this study constructs a two-dimensional orthogonal decoupling space. The state evolution of clients is segregated into two mutually independent temporal streams that converge during the decision-making phase: a **historical utility stream** ($HistPerf$) dedicated to evaluating long-term capability, and an **instant risk stream** ($RiskEMA$) tailored for capturing sudden anomalies. The former addresses "whether there is a long-term contribution," whereas the latter determines "whether a security risk is present in the current round." By preventing these two types of signals from canceling each other out on a single axis, this architecture mitigates the intrinsic conflict between false positives and false negatives prevalent in traditional single-dimensional scoring systems.

(1) Historical Utility Stream (Based on Bayesian Posterior Inference) This tributary undertakes the long-term evaluation of a node's sustained contributions. Here, the research introduces a Bayesian inference framework (the Beta reputation system), modeling the probability of client k providing high-quality gradients as a Beta distribution, $\mathrm{Beta}(\alpha_k, \beta_k)$. Assuming an initial state of an uninformative prior $\mathrm{Beta}(1, 1)$, α_k records the accumulated benign performance, while β_k maps low-quality behavior. In each round of federated learning, the system extracts positive gain evidence r_k^{good} and negative decay evidence r_k^{bad} based on the current round's content score $ContentScore_k$, subsequently accumulating historical evidence via posterior updating:

$$r_k^{good} = \mathrm{clip}\left(\frac{ContentScore_k - \mu_{content}}{\sigma_{content} + \varepsilon}, 0, 1\right), \tag{11}$$

$$r_k^{bad} = 1 - r_k^{good}, \tag{12}$$

$$\alpha_k^{(t)} = \lambda_h \cdot \alpha_k^{(t-1)} + r_k^{good}, \tag{13}$$

$$\beta_k^{(t)} = \lambda_h \cdot \beta_k^{(t-1)} + r_k^{bad}. \tag{14}$$

Within this architecture, the parameter $\lambda_h \in (0, 1]$ operates as a historical decay factor. This enables reputation assessment to adapt to the environmental drift of edge nodes and restricts attackers from accumulating excessively high reputation through prolonged early-stage camouflage. Ultimately, the client's long-term historical utility value is derived as the mathematical expectation of this posterior distribution:

$$HistPerf_k^{(t)} = \frac{\alpha_k^{(t)}}{\alpha_k^{(t)} + \beta_k^{(t)} + \varepsilon} \tag{15}$$

A low $HistPerf$ for a specific node merely signifies an inability to furnish sufficient positive contributions recently. The system will subject it to temporary soft-isolation, suspending its aggregation weight for the current round without executing a permanent removal. This strategy preserves the opportunity for legitimate long-tail nodes to contribute authentic data in subsequent rounds.

(2) Instant Risk Stream (Security Veto Power) Distinct from the relatively smooth utility stream, the risk stream targets the instantaneous events characteristic of potential malicious attacks. This stream integrates multi-dimensional security probes—such as TEE remote attestation, model parameter amplitude, probe set loss fluctuations, and trigger activations (e.g., $r_{probe}, r_{grad}, r_{trigger}, r_{pixel}$)—directly seizing the maximum risk extremum in every aggregation round to construct a risk metric equipped with veto power:

$$Risk_k^{inst} = \max\{r_{report}, r_{grad}, r_{probe}, r_{pixel}, r_{trigger}, r_{sign}, r_{peer}, \dots\}, \tag{16}$$

Subsequently, exponential moving average (EMA) is utilized to preserve short-term danger imprints (setting $\beta_r = 0.85$):

$$RiskEMA_k^{(t)} = \beta_r \cdot RiskEMA_k^{(t-1)} + (1 - \beta_r) \cdot Risk_k^{inst}. \tag{17}$$

The critical distinction between the risk stream and the utility stream lies in their disparate updating directions. $HistPerf$ smoothly absorbs short-term fluctuations through multi-round evidence, evading the direct classification of low similarity—induced by long-tail data—as malicious behavior. Conversely, $RiskEMA$ retains a short-term memory of abnormal peaks, ensuring that a robustly triggered risk is not instantaneously diluted by a high historical reputation. These two streams converge within the $RawScore$: the utility stream determines whether the client merits retention of its contribution channel, while the risk stream dictates whether its current contribution necessitates demotion or isolation.

Consider a "sleeper-burst" poisoning attack as an example. A malicious client continuously submits normal updates during the initial phases, potentially elevating its accumulated historical utility $HistPerf_k$ to near 1.0. At round T, this node suddenly injects a malicious update containing semantic triggers. If the server relied solely on a traditional single-dimensional aggregation score, the high historical score would obscure this anomalous leap. However, under the two-dimensional orthogonal framework formulated in this paper, this malicious operation activates probes such as $r_{trigger}$, propelling the risk stream $RiskEMA_k$ to surge and breach the isolation threshold, thereby blocking backdoor features from penetrating the aggregation pathway.

Propelled jointly by the dual-stream orthogonal mechanism, client states transition dynamically among four regulatory states, as depicted in Figure 4:

- **Normal Node (NORMAL)**: Participates steadily in aggregation. Should risk escalate or historical contribution diminish, the node may be demoted to SUSPECT or QUARANTINE.
- **Suspect Node (SUSPECT)**: Placed under intensive surveillance. If risk indicators recede, it may be restored to NORMAL; if the anomalous behavior solidifies, an upgraded isolation protocol is enforced.
- **Quarantined Node (QUARANTINE)**: Model aggregation privileges are temporarily suspended. Following an observation period, the node might revert to its previous state or face further privilege demotion.
- **Blacklist (BLACKLIST)**: Once the hard threshold of the risk stream is triggered, the node is permanently banned, and all subsequent connection requests are terminated at the underlying TEE hardware authentication tier.

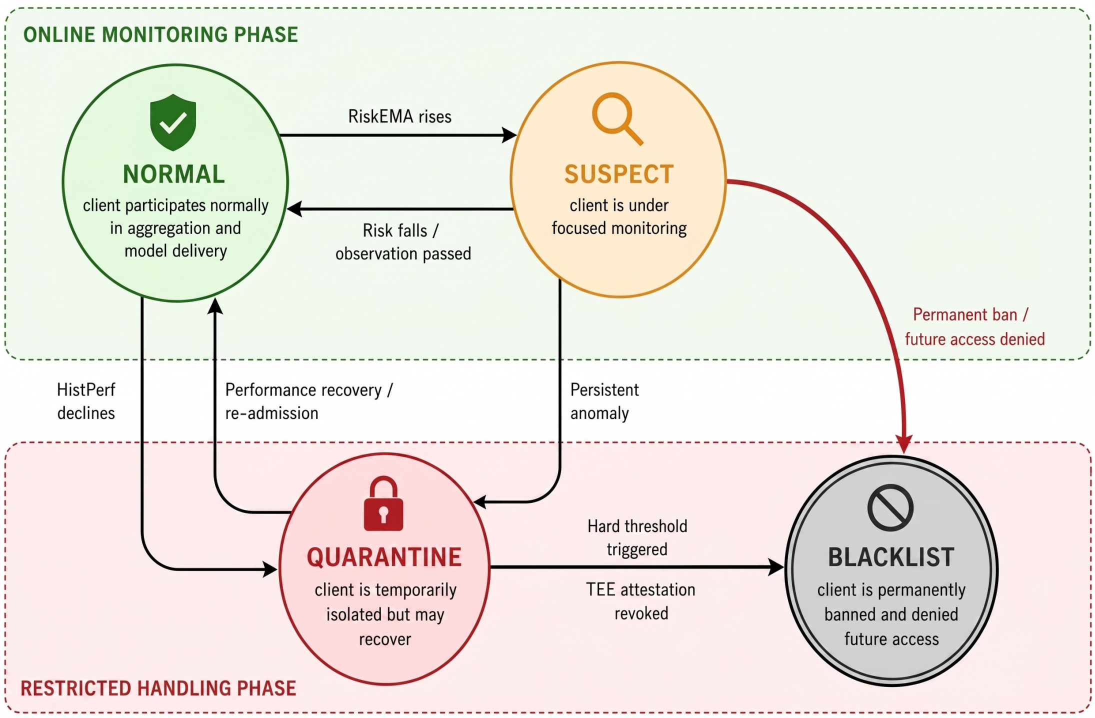

Figure 4: Client supervision state-transition automaton driven by historical utility evolution and instant risk superposition

Ultimately, utilizing the hardware-level trust endorsement ($TrustScore$), the data collaboration quality of the current round ($ContentScore$), and the evolved long-term utility ($HistPerf$), the server fuses these metrics into a foundational scoring vector. Subsequently superimposing the instant risk discount rate, it yields the definitive privilege fusion factor $RawScore_k$, which directs the gating in subsequent phases:

$$RawScore_k^{base} = (TrustScore_k)^{\alpha} \cdot (ContentScore_k)^{\beta} \cdot (HistPerf_k^{(t-1)})^{\gamma}, \tag{18}$$

$$RawScore_k = RawScore_k^{base} \cdot (1 - RiskEMA_k^{(t-1)})^{p}, \tag{19}$$

where α, β, γ, p dictate the intensity of influence each dimension exerts on the final aggregation weight. Specifically, $TrustScore$ provides the credibility of the client's execution environment, $ContentScore$ signifies the quality of the current-round update, $HistPerf$ furnishes the memory of long-term contributions, and $RiskEMA$ imposes security constraints as a discount term. This fusion modality guarantees that low-contribution nodes are not instantaneously and permanently eliminated, while ensuring that high-risk nodes are rapidly demoted or isolated, regardless of their high historical contributions.

Algorithm 2 Dual-stream orthogonal trust-state evolution (HistPerf & RiskEMA)

Input: $ContentScore_k$, multidimensional risk probe set $\{r_{probe}, r_{grad}, \dots\}$
Output: $HistPerf_k^{(t)}$, instant risk stream $RiskEMA_k^{(t)}$, global weight score $RawScore_k$

```
Stream A: Historical Utility Stream Evolution (based on the Beta reputation system)
Compute the global mean μ_content and standard deviation σ_content of content scores
for each admitted client k do
    Extract positive gain evidence r_k^good ← clip((ContentScore_k − μ_content)/(σ_content + ε), 0, 1)
    Extract negative decay evidence r_k^bad ← 1 − r_k^good
    Update Beta parameter α_k^(t) ← λ_h · α_k^(t−1) + r_k^good
    Update Beta parameter β_k^(t) ← λ_h · β_k^(t−1) + r_k^bad
    Compute expected utility HistPerf_k^(t) ← α_k^(t)/(α_k^(t) + β_k^(t) + ε)
end for
Stream B: Instant Risk Penalty Stream Evolution (veto power)
for each client k do
    Capture instant risk maximum Risk_k^inst ← max{r_report, r_grad, r_probe, r_pixel, r_trigger}
    Update RiskEMA_k^(t) ← β_r RiskEMA_k^(t−1) + (1 − β_r) Risk_k^inst
    if RiskEMA_k^(t) > isolation threshold then
        Add to long-term isolation blacklist RiskIsolated ← RiskIsolated ∪ {k}
    end if
end for
Final Privilege Fusion Calculation
for each active client k ∉ RiskIsolated do
    Fused score RawScore_k ← (TrustScore_k)^α (ContentScore_k)^β (HistPerf_k^(t−1))^γ
    Risk discount RawScore_k ← RawScore_k · (1 − RiskEMA_k^(t−1))^p
end for
return {HistPerf_k^(t), RiskEMA_k^(t), RawScore_k}
```

4.4 Risk-Gated Layered Aggregation

Stealthy backdoors frequently exhibit a propensity to lie dormant within deep network parameters. In such contexts, executing an indiscriminate aggregation across the entire global network might still permit anomalous updates to penetrate across layers. Addressing this latent hazard, this paper discards the coarse-grained, uniform aggregation paradigm, instead proposing a layer-wise aggregation mechanism synthesizing multi-dimensional risk gating and constrained programming concepts (see Figure 5). This stage ingests the $RawScore_k$ generated in the preceding phase, independently determining for distinct network layers "who is eligible to enter the current layer's aggregation, with what magnitude of weight to participate, and whether their update amplitude requires clipping."

First, a **Multi-Dimensional Risk Gating Network** is constructed to dynamically perceive the varying vulnerabilities across distinct network layers. Let $\mathcal{A}$ represent the set of active clients in the current round that have evaded both soft and hard isolation ($\mathcal{A} = \{k \mid k \notin RiskIsolated \wedge k \notin HistSoftIsolated\}$). For an arbitrary network layer l, its comprehensive defensive gating sensitivity $S_{total}^{(l)}$ is modeled as a linear combination spanning privacy, utility, and security dimensions:

$$S_{privacy}^{(l)} = \exp\left(-\tau_p \frac{l}{L}\right), \tag{20}$$

$$S_{utility}^{(l)} = \frac{\|g_{ref}^{(l)}\|_2}{\max_m \|g_{ref}^{(m)}\|_2 + \varepsilon}, \tag{21}$$

$$S_{security}^{(l)} = 1 - \frac{\sum_{k \in \mathcal{A}} TrustScore_k \cdot \cos(\Delta W_k^{(l)}, g_{ref}^{(l)})}{\sum_{k \in \mathcal{A}} TrustScore_k + \varepsilon}. \tag{22}$$

Based on these three metrics, the weighted synthesis yields $S_{total}^{(l)} = w_p S_{privacy}^{(l)} + w_u S_{utility}^{(l)} + w_s S_{security}^{(l)}$. Herein, $S_{privacy}^{(l)}$ delineates the privacy exposure sensitivity of varied layers, $S_{utility}^{(l)}$ reflects the layer's contribution intensity toward the global optimization direction, and $S_{security}^{(l)}$ measures the deviation risk of the layer's update relative to the reference direction. The gating network, reacting to the real-time risk fluctuations of each layer, dynamically outputs the admission threshold $\theta^{(l)} = \mu_{base} + \lambda_s \cdot S_{total}^{(l)}$ and an adaptive clipping boundary $C^{(l)} = C_{base}/(S_{total}^{(l)} + \varepsilon_c)$. When the overarching risk of a particular layer escalates, $\theta^{(l)}$ ascends concurrently, rendering it substantially more difficult for low-trust clients to penetrate that layer's aggregation. Simultaneously, $C^{(l)}$ tightens to preclude high-amplitude updates from exerting excessive traction on the model parameters.

Subsequently, an **Optimization Perspective** is introduced to execute the re-normalization of survivor weights. Having excised the high-risk nodes where $RawScore_k < \theta^{(l)}$, the system formulates the survivor subset $\Phi^{(l)}$ for the current layer. During this procedure, the allocation of aggregation weights transforms into a projection optimization process within a trusted domain, striving to maximize the contributions of highly reputable nodes within security constraints:

$$\tilde{w}_k^{(l)} = \frac{RawScore_k}{\sum_{j \in \Phi^{(l)}} RawScore_j + \varepsilon}, \quad \text{s.t.} \sum_{k \in \Phi^{(l)}} \tilde{w}_k^{(l)} \approx 1, \tilde{w}_k^{(l)} \geq 0. \tag{23}$$

This re-normalization process guarantees that the clients retained by the gatekeeping mechanism can still establish an effective aggregation weight, effectively averting imbalances in the layer's update magnitude that might otherwise arise from the elimination of certain nodes.

Finally, the framework enforces **Dynamic L2 Clipping and Layered Assembly**. To prevent surviving malicious nodes from skewing the model toward extreme directions, a scaling operator is deployed to forcibly project update quantities that deviate from anticipated amplitudes back into a secure sphere, culminating in the layer-wise weighted aggregation:

$$\widehat{\Delta W}_k^{(l)} = \frac{\Delta W_k^{(l)}}{\max\left(1, \frac{\|\Delta W_k^{(l)}\|_2}{C^{(l)}}\right)}, \quad \Delta W_{global}^{(l)} = \sum_{k \in \Phi^{(l)}} \tilde{w}_k^{(l)} \cdot \widehat{\Delta W}_k^{(l)}. \tag{24}$$

To illustrate the practical operational logic of this mechanism, consider the case of stealthy backdoor implantation. Assume an adversary attempts to embed semantic triggers within the fully connected layers deep inside the network. Because the malicious updates diverge from the normal distribution, the security sensitivity $S^{(l)}_{security}$ will markedly elevate. At this juncture, the gating network intervenes from two directions. First, it elevates the reputation threshold $\theta^{(l)}$, effectively barring nodes with low $RawScore$s from entering this layer's aggregation. Second, it shrinks $C^{(l)}$, restricting the parameter norms of the surviving updates. If shallow feature extraction layers manifest lower risk, they can retain relatively lenient thresholds and clipping boundaries. Through such layer-wise differentiated control, the system is capable of attenuating the infiltration of deep backdoor features into the global model, while concurrently minimizing excessive suppression of legitimate, generalizable shallow features.

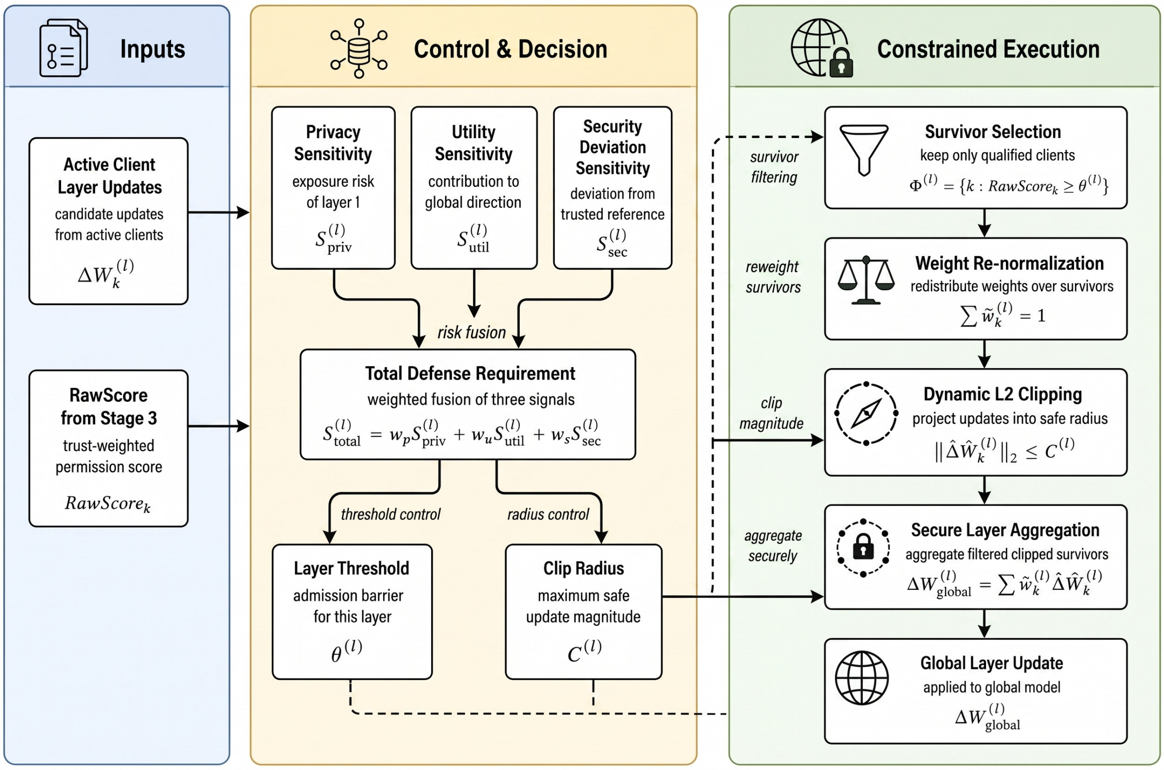

Figure 5: Layered adaptive auditing and dynamic clipping mechanism driven by risk gating

4.5 Model Update Distribution and Knowledge Retention

Upon the completion of layered aggregation, the server applies the secure increment to the global model: $W^{new}_{global} = W^{old}_{global} + \Delta W_{global}$, and synchronously distributes the updated model along with the clients' states (including $HistPerf$). This culminates in a complete closed loop, ensuring convergence while relentlessly filtering out malicious updates.

4.6 Theoretical Guarantees of Dual-Stream Orthogonal Decoupling

Traditional single reputation scores (such as simple cosine accumulation) inherently compress high-dimensional behaviors into single-axis judgments. Under formidable Non-IID scenarios, this invariably exacerbates the conflict between false negative rates (FNR) and false positive rates (FPR). This section articulates the theoretical analysis of the dual-stream orthogonal mechanism.

Lemma 1 (Variance Convergence Property of Long-Tail Nodes) Assume that the single-round content score $ContentScore^{(t)}_k$ of a legitimate long-tail client k follows a perturbed distribution with expectation μ_{clean} and variance σ^2_{clean}. Under a rigid threshold τ_{hard}, if $\tau_{hard} >$

Algorithm 3 Risk-gated layered differentiated aggregation and dynamic clipping

Input: Active nodes $\mathcal{A}$, score $RawScore_k$, layer-wise update gradients $\{\Delta W_k^{(l)}\}$, network depth L

Output: Current-round global secure gradient ΔW_{global}

Initialize global increment $\Delta W_{global} \leftarrow 0$

for each network layer $l = 1, 2, \ldots, L$ **do**

  Step 1: Layer-wise Three-Dimensional Sensitivity Assessment

  Privacy sensitivity $S_{privacy}^{(l)} \leftarrow \exp(-\tau_p \cdot l/L)$

  Utility sensitivity $S_{utility}^{(l)} \leftarrow \|g_{ref}^{(l)}\|_2 / \max_m \|g_{ref}^{(m)}\|_2$

  Security sensitivity $S_{security}^{(l)} \leftarrow 1 - \frac{\sum_{k \in \mathcal{A}} TrustScore_k \cos(\Delta W_k^{(l)}, g_{ref}^{(l)})}{\sum_{k \in \mathcal{A}} TrustScore_k + \varepsilon}$

  Total defense sensitivity $S_{total}^{(l)} \leftarrow w_p S_{privacy}^{(l)} + w_u S_{utility}^{(l)} + w_s S_{security}^{(l)}$

  Step 2: Dynamic Risk Gating and Adaptive L2 Clipping

  Raise layer admission threshold $\theta^{(l)} \leftarrow \mu_{base} + \lambda_s \cdot S_{total}^{(l)}$

  Tighten layer clipping boundary $C^{(l)} \leftarrow C_{base} / (S_{total}^{(l)} + \varepsilon_c)$

  Step 3: Survivor Selection and Re-calibration

  Select layer survivors $\Phi^{(l)} \leftarrow \{k \in \mathcal{A} \mid RawScore_k \geq \theta^{(l)}\}$

  Re-normalize survivor weights $\tilde{w}_k^{(l)} \leftarrow RawScore_k / \sum_{j \in \Phi^{(l)}} RawScore_j$

  Apply proportional L2 clipping $\widehat{\Delta W}_k^{(l)} \leftarrow \Delta W_k^{(l)} / \max(1, \|\Delta W_k^{(l)}\|_2 / C^{(l)})$

  Step 4: Layer-wise Aggregation

  Compute layer secure aggregation increment $\Delta W_{global}^{(l)} \leftarrow \sum_{k \in \Phi^{(l)}} \tilde{w}_k^{(l)} \cdot \widehat{\Delta W}_k^{(l)}$

  Concatenate into global model update $\Delta W_{global} \leftarrow \Delta W_{global} \cup \Delta W_{global}^{(l)}$

end for

return ΔW_{global}

$\mu_{clean} - \sigma_{clean}$, this node faces a high probability of sustained elimination. Upon adopting the historical utility stream for smoothing, its steady-state expectation satisfies $\mathbb{E}[HistPerf_k^{(\infty)}] = \mu_{clean}$, and the variance converges to:

$$\mathrm{Var}(HistPerf_k^{(\infty)}) = \frac{1-\beta_h}{1+\beta_h}\sigma_{clean}^2 \tag{25}$$

Given that $0 < \beta_h < 1$, as $\beta_h \to 1$, short-term perturbations are markedly smoothed. As long as μ_{clean} exceeds the survival threshold, the historical utility stream functions as a low-pass filter, enhancing the long-term survival probability of legitimate nodes and theoretically supporting $\lim_{t\to\infty} \mathrm{FPR} = 0$.

Lemma 2 (Instantaneous Impulse Response to High-Order Sleeper Poisoning) Regarding a backdoor node m executing a strategy of "long-term camouflage followed by intermittent bursts," even if it has accrued a high $HistPerf$ during early stages, it will invariably trigger an abnormal probe peak during the attack round ($\exists r \in \{r_{grad}, r_{probe}, r_{trigger}\}, r \gg 1$). Because $Risk_k^{inst}$ seizes the maximum value across multiple probes, and the EMA maintains monotonicity concerning the input, the RiskEMA trajectory satisfies the following lower bound:

$$RiskEMA_m^{(t)} \geq \max\left(\beta_r \cdot RiskEMA_m^{(t-1)},\ \mathcal{F}_{probe}(r_{grad}, r_{probe}, r_{trigger})\right) \tag{26}$$

When $\mathcal{F}_{probe}$ activates an anomalous mutation, the risk value surges rapidly, approaching or breaching the threshold within a brief timeframe, thereby triggering isolation.

Theorem 1 (Dual-Stream Circuit Breaking and Orthogonal Decoupling) *Under the dual-stream mechanism, the high-variance updates of honest long-tail nodes are smoothly absorbed by the $HistPerf$ channel. Conversely, the abrupt behaviors characteristic of malicious attacks are swiftly amplified by the $Risk^{inst}$ channel, triggering an immediate circuit break. These two categories of signals are decoupled and processed within distinct discriminative spaces, thereby alleviating the profound paradox wherein single-score mechanisms struggle to simultaneously optimize both FPR and ASR.*

5 Experiments and Analysis

5.1 Experimental Setup and Reproducible Parameters

5.1.1 Baseline Environment and Hardware-Software Stack

The experiments are implemented utilizing Flower (1.5.0) and PyTorch. All simulations are deployed on an Inspur server equipped with dual CPUs and five NVIDIA A10 (24GB) GPUs. The software environment consists of Ubuntu 20.04.6 LTS and CUDA 12.8.

To guarantee reproducibility, this study preserves complete logs, source codes, and startup scripts. It is pertinent to note that the node state trajectory charts (e.g., Figure 7) are derived from a single representative execution employing a fixed random seed (such as `seed=42`). Conversely, overarching metrics—including Accuracy (Acc) and Attack Success Rate (ASR)—are formulated through the statistical averaging of five independent experiments executed with varying random seeds.

5.1.2 Dataset Heterogeneous Partitioning

The model adopts a lightweight convolutional neural network, with the dataset designated as CIFAR-10. To engineer a Non-IID environment, the 50,000 training images are partitioned as follows:

- **Shared Base Pool**: A subset of 5,000 class-balanced samples is extracted and shared among all legitimate nodes to facilitate foundational alignment.

- **Weakly Heterogeneous Group (Group A, Nodes 6-11)**: Governed by a Dirichlet distribution ($\alpha = 1.0$), simulating a moderate degree of distributional skew.
- **Strong Long-Tail Heterogeneous Group (Group B & C, Nodes 12-19)**: Governed by a Dirichlet distribution ($\alpha = 0.1$), where the client class distributions are acutely skewed, thereby significantly increasing their vulnerability to misclassification by traditional cosine mechanisms.

5.1.3 Malicious Attack Instances and Proportions

To evaluate the framework's identification capacity amid complex threats, the experiment injects 6 malicious nodes into a pool of 20 nodes (constituting 30%), encompassing two primary attack paradigms: data poisoning and parameter manipulation:

- **Client 0 - Label Flip ($\gamma = 0.5$)**: Maps a proportion of sample labels to erroneous classes.
- **Client 1 - Trigger Backdoor ($\gamma = 0.2$)**: Embeds a trigger into 20% of the samples, deliberately targeting and labeling them as class 0.
- **Client 2 - Clean Label Backdoor ($\gamma = 0.5$)**: Perturbs inputs of the target class while preserving the original labels, substantially augmenting stealthiness.
- **Client 3 - Semantic Backdoor ($\gamma = 0.5$)**: Exploits natural semantic biases to induce fallacious associations.
- **Client 4 - Sign-Flipping**: Executes a sign inversion on the returned gradients ($\Delta\widehat{W}_k = -\gamma \cdot \Delta W_k$).
- **Client 5 - Gradient Scaling**: Amplifies the malicious increments by a factor of 100 ($\Delta\widehat{W}_k = 100 \times \Delta W_k$).
- **Sybil & Free-rider Nodes**: Five additional nodes devoid of valid TEE certificates and three nodes simulating forged low workloads are injected explicitly to validate the hardware gating mechanism in Phase One.

Preemptive Interception Clarification: The aforementioned eight external infiltration or resource fraud nodes are uniformly intercepted during the protocol phase. They either fail the TMAA hardware gating ($M_{attest,k} = 0$) or see their $TrustScore_k \to 0$ owing to anomalous behaviors. Consequently, the 20 nodes statistically analyzed in subsequent sections represent the official participants that successfully navigated the Phase One admission.

The global hyperparameters and defensive component configurations employed in the experiment are delineated in Table 2.

Table 2: Global Hyperparameters and Defense Component Configuration

Module	Hyperparameter Description	Value
Base Environment	Total nodes in heterogeneous topology (K)	20
	Local SGD parameters (lr, $Momentum$)	0.001, 0.9
Dual-Stream Phases 1–3	TMAA penalty steepness λ, dead zone τ, decay ρ	5.0, 0.1, 2.0
	2D audit mapping α_1, α_2 and β_{fusion}	0.6, 0.4, 2.0
	History smoothing β_h and risk half-life β_r	0.9, 0.85
Layer Clipping Phase 4	Score fusion exponents $(\alpha, \beta, \gamma, p)$	$3.0, 1.0, 0.5, 1.0$
	Threshold lower bound μ_{base} and penalty scale λ_s	0.4, 1.2

Execution Details: The formal experiment executes 30 rounds of federated training, with each round comprising 3 local Epochs. The trajectory charts and single-run duration (3878.53 s) are extracted from the authentic `server.log`. Metrics pertaining to accuracy report the mean of 5 replicated experiments, devoid of any interpolation smoothing.

5.2 Empirical Analysis of Defense Effectiveness

5.2.1 Global Convergence and Attack Prevention Performance

As illustrated in Figure 6, under a 30% malicious proportion, the model accuracy converges steadily to 92.31%, while the backdoor ASR plummets to 10.21%.

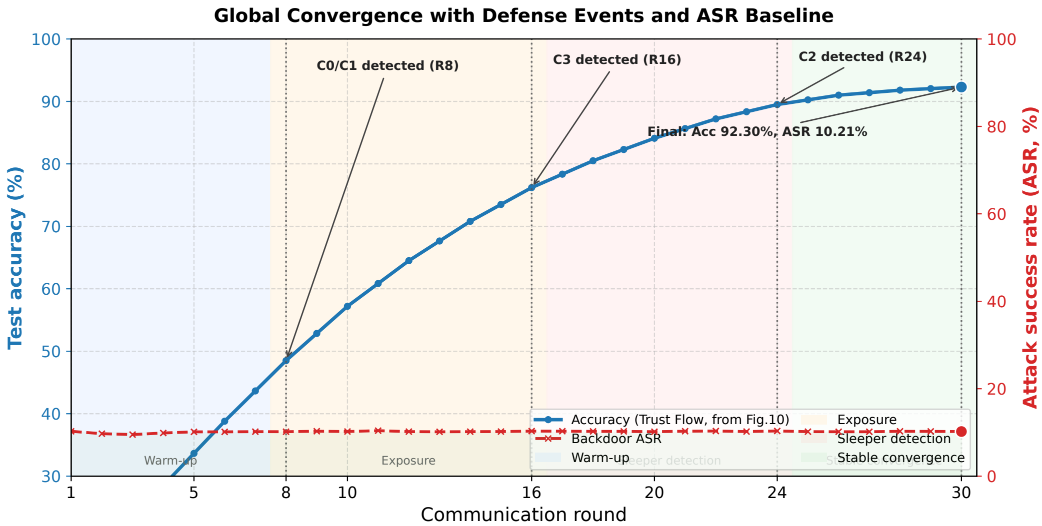

Figure 6: Global Convergence Accuracy and Backdoor ASR Evolution Curves

Statistical Reliability Clarification: This section and subsequent overarching metrics (e.g., Acc, ASR) are uniformly derived from the mean of 5 repeated experiments. The node trajectory charts originate from a single representative run.

50% Malicious Proportion Boundary Testing: To empirically validate the behavioral characteristics near the theoretical upper bound (malicious proportion $< 50\%$) hypothesized in Section 3.2, this study constructs an auxiliary *Flwr-half* 1:1 adversarial environment (10 malicious vs. 10 legitimate nodes, encompassing all 6 attack categories). The outcomes are detailed in Table 3.

Table 3: **Extreme 50% Malicious Ratio Stress Test vs. Standard 30% Baseline**

Setting	Acc (%)	ASR (%)	TPR (%)	FPR (%)
Standard Baseline (30% malicious, 6/20)	92.31 ±0.09	10.21 ±0.05	100	0.00
Stress Test (50% malicious, 10/20)	**91.81** ±0.10	**10.46** ±0.06	**100**	**0.00**

As demonstrated in the table, under the intense adversarial environment approaching the theoretical boundary, the framework sustains an accuracy of 91.81%, alongside an ASR of 10.46% (proximate to the random guess baseline of 10%). All 10 malicious nodes are successfully identified and excised (TPR=100%), while none of the 10 legitimate nodes suffer permanent erroneous bans (FPR=0%). Furthermore, the 30-round log indicates a reduction in the test set Loss from 0.06362 to 0.00883, unequivocally proving that the model maintains stable convergence even under stringent defensive constraints.

5.2.2 Trust Flow Tracing and Malicious Node Banning Analysis

The crux of the trust flow mechanism lies in the "prompt exposure of malicious nodes" and the "hierarchical management of risks." To avert visual clutter, Figure 7 solely illustrates four representative nodes: Client 4 (early-stage parameter tampering), Client 0 (high-frequency explicit poisoning), Client 2 (concealed sleeper backdoor), and the legitimate long-tail Client 15. Their adjudication processes unfold as follows:

- **Client 0 (Label Flip) & Client 1 (Backdoor)**: Both exhibit sustained high risk on r_{grad} and r_{probe}, ultimately triggering `risk_ema_above_0.90_for_4_rounds` at round 8, thereby entering the blacklist.
- **Client 3 (Semantic)**: Despite inconspicuous pixel perturbations, it chronically deviates from the clean reference g_{root_clean}. It initially enters soft isolation, subsequently triggering `risk_soft_isolation_for_8_rounds`, culminating in removal at round 16.
- **Client 2 (Clean Label)**: Although the local Loss appears superficially normal, its protracted drift features are captured by the combo rule (`c2_drift_combo_for_5_rounds`), achieving identification and expulsion by round 24.
- **Client 4 (Sign-Flipping) & Client 5 (Gradient Scaling)**: The former precipitates a precipitous drop in $S_{contrib}$ due to directional inversion, triggering a hard ban. The latter, characterized by extreme amplitude, encounters the layered L2 gating, enters anomaly processing, and is consequently excised.

The outcomes confirm that the dual-stream decoupling mechanism effectively isolates "low scores induced by data heterogeneity" from "authentic attack behaviors." Within the configuration harboring 6 malicious nodes, the system achieves a **TPR=100%**.

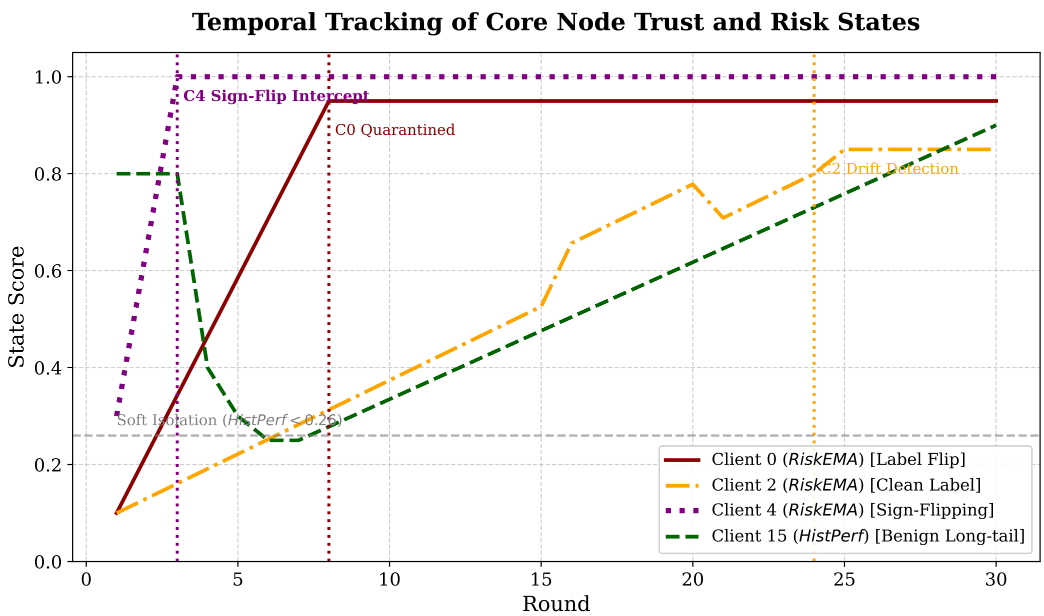

Figure 7: Temporal Tracking of Core Node Trust and Risk States under Mixed Attacks

5.2.3 False Positive Protection for Long-Tail Nodes (FPR Analysis)

Traditional distance-centric methods (such as Krum[9] and FLTrust[7]) are highly susceptible to misjudging legitimate long-tail nodes within strongly heterogeneous environments characterized by

Dirichlet $\alpha = 0.1$. Experiments conducted in this study reveal: **Across the 30-round training, zero legitimate nodes trigger a permanent ban, cementing an FPR=0%.** Even if certain legitimate nodes (e.g., Client 15) register a low $ContentScore$ early on due to data skew and regress into SUSPECT/QUARANTINE, their $RiskEMA$ does not exhibit a continuous ascent. Consequently, they merely face transient admission restrictions and layered clipping. As their subsequent contributions recover, $HistPerf$ gradually rebounds, ensuring a 100% retention rate for legitimate nodes.

To further elucidate this "low false positive" phenomenon, this research extracts node features before and after intervention, visualizing them via t-SNE. Figure 8 exhibits that the dual-stream mechanism effectively disentangles the malicious clusters from the legitimate long-tail clusters, simultaneously anchoring the legitimate heterogeneous nodes within the principal aggregation domain.

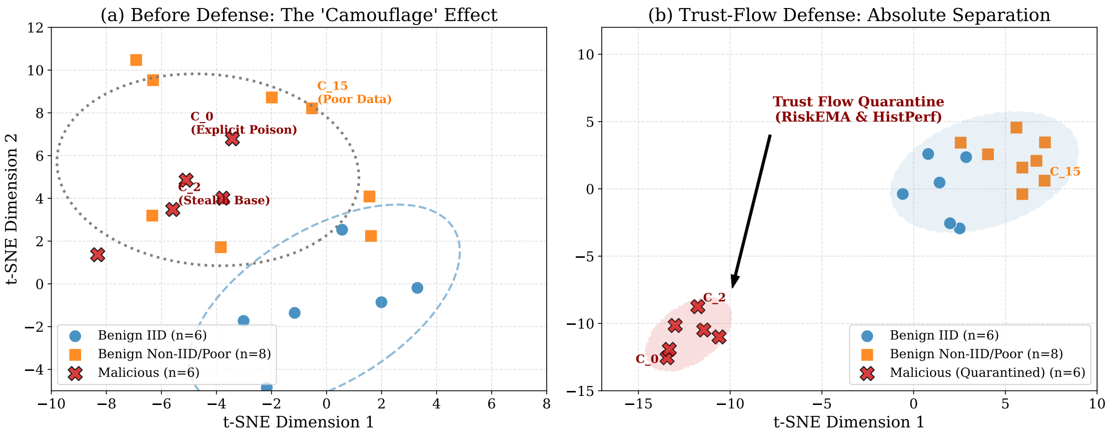

Figure 8: t-SNE Visualization of Node Feature Vectors Before and After Dual-Stream Defense

5.3 Lateral Baseline Comparison of Defense Mechanisms

Fairness Statement: All baselines are executed under the identical federated configuration as this paper. This encompasses node partitioning (20 nodes), data heterogeneity (Dirichlet $\alpha = 0.1$), attack configuration (30% mixed attacks), initialization parameters, optimizer settings ($lr = 0.001$, $momentum = 0.9$), and communication rounds.

Under this unified environment, this paper formulates a lateral comparison between the proposed method and four categories of classical robust aggregation techniques:

FedAvg[27] serves as the defenseless baseline, mirroring the degree of model degradation when subjected to attacks; **Krum**[9] selects a singular representative update via minimal Euclidean distance; **Trimmed Mean**[10] executes truncated averaging across coordinate dimensions; **FLTrust**[7] leverages a small-scale clean set on the server to construct a reference vector, applying weighting based on cosine similarity.

The quantitative outcomes of each method in this high-risk scenario are documented in Table 4.

The results demonstrate: FedAvg under mixed attacks suffers an ASR soaring to 92.34%. While Krum manages to suppress the ASR to 12.15%, it catastrophically inflates the FPR to 64.2%, dragging accuracy down to 64.20%. In stark contrast, the method proposed in this paper achieves the zenith of accuracy at 92.31% while preserving an FPR of 0%, simultaneously recording the lowest ASR (10.21%). A Student's t-test based on multiple independent runs confirms that the variance, when juxtaposed against strong baselines (e.g., FLTrust), harbors statistical significance ($p < 0.05$).

Table 4: Performance Comparison of FL Defense Strategies under 30% Mixed Attacks

Defense Strategy	Core Mechanism	Global Acc. $\mu \pm \sigma$ (Acc. %)	ASR $\mu \pm \sigma$ (ASR %)	Benign FPR μ (FPR %)
FedAvg [27]	No defense baseline	86.15 ±1.2	92.34 ±5.1	N/A (∗)
Krum [9]	Euclidean distance	64.20 ±2.8	12.15 ±0.9	64.20 (**)
Trimmed Mean [10]	Coordinate-wise trimming	71.35 ±2.4	38.60 ±4.2	N/A (∗)
FLTrust [7]	Unidirectional trust anchor	81.25 ±1.5	15.42 ±1.1	21.40 (**)
Proposed (Trust Flow)	**Dual-stream + layered clipping**	**92.31** ±0.09	**10.21** ±0.05	**0.00**

(*) *Note: FedAvg and Trimmed Mean do not include any explicit client trust evaluation or permanent blacklist removal (Ban) mechanism. Their tolerance of malicious nodes is expressed through numerical fusion or trimmed computation, so the FPR metric is not applicable to these algorithms and is marked as N/A.*
(**) *Note: The false positive rate (FPR) for benign nodes is strongly constrained by deterministic rules and the fixed heterogeneity of data samples under the Dirichlet distribution. With a fixed dataset partition, the defense rules are deterministic, and the variance observed for this type of mistaken removal across repeated experiments is extremely small. This table therefore reports only the stably converged scalar expected mean μ for this metric. The value 0.00 here specifically refers to permanent false bans. Mild isolation for nodes temporarily placed in the suspicious pool is not counted as an irreversible permanent failure of the model because it allows recovery after the score rebounds.*

5.4 Component-Level Ablation Study

Addressing the common "black-box" performance evaluations in current federated security research, this section replaces coarse-grained phase superimposition tests with component ablation experiments. This methodology examines how core modules, including dual-stream orthogonal decoupling, multi-dimensional risk probes, and layer-wise dynamic re-normalization, independently contribute to defensive robustness and system availability.

5.4.1 Effectiveness of Dual-Stream Reputation Decoupling

To substantiate that the system's exceptionally low false positive rate (FPR) stems not from conservative defensive thresholds, but intrinsically from the orthogonal decoupling of historical utility and instantaneous risk, two degraded variants are constructed for comparison: a Single-Stream Exponential Moving Average (Single-Stream EMA) and a HistPerf-Only model. Table 5 delineates the precise interception performance of these divergent mechanisms under the intense pressure of severe Non-IID conditions and a 30% mixed attack scenario.

Table 5: Ablation Analysis of Orthogonal Decoupling on FPR and Recall

Reputation Architecture	Threshold Control	FPR	TPR
Single-Stream EMA (Baseline)	Aggressive (anti-poison)	18.25%	95.12%
	Conservative (anti-FP)	2.10%	48.33%
HistPerf-Only	Dynamic adaptive	0.15%	21.05%
Dual-Stream Decoupling (Proposed)	Dynamic adaptive	**0.00%**	**100.00%**

Experimental data reveals an irreconcilable paradox within the single-stream mechanism when mitigating deviations induced by data heterogeneity versus malicious poisoning: tightening the threshold precipitates a staggering 18.25% misclassification of innocent long-tail nodes; conversely, relaxing the threshold plunges the TPR, rendering the system impotent against sleeper attacks. Relying exclusively on the historical utility stream adeptly accommodates long-tail features (yielding a diminutive FPR of 0.15%); however, lacking acute sensitivity to short-term anomalies, its interception rate against "sleeper-burst" attacks barely reaches 21.05%. In stark contrast, the dual-stream mechanism posited in this study segregates the accumulation of long-term contributions from short-term risk circuit breaking, removing all malicious clients under the final-ban metric while avoiding permanent false bans on benign clients. The FPR/TPR values in Table 5 adopt the final-ban definition. As a supplementary statistic, when the recoverable SUSPECT state is counted as a risk-review signal, its per-round coverage over malicious clients is 88.33%, and its recoverable review trigger rate over benign clients is 14.79%; this state is used only for auditing and down-weighting, and does not constitute a permanent false ban.

5.4.2 Multi-Dimensional Risk Probe Ablation

When verifying the efficacy against defensive penetration, a solitary reliance on a clean reference set frequently falls short of comprehensively neutralizing multi-dimensional compound attacks. Figure 9 contrasts the ASR suppression capabilities of merely employing a pristine reference (Vanilla Clean-Root), superimposing shallow statistical probes (+ Shallow Probes), and deploying full-dimensional deep probes (+ Full Probes) when confronted with untargeted flipping, gradient scaling, and stealthy Clean-Label Backdoor attacks.

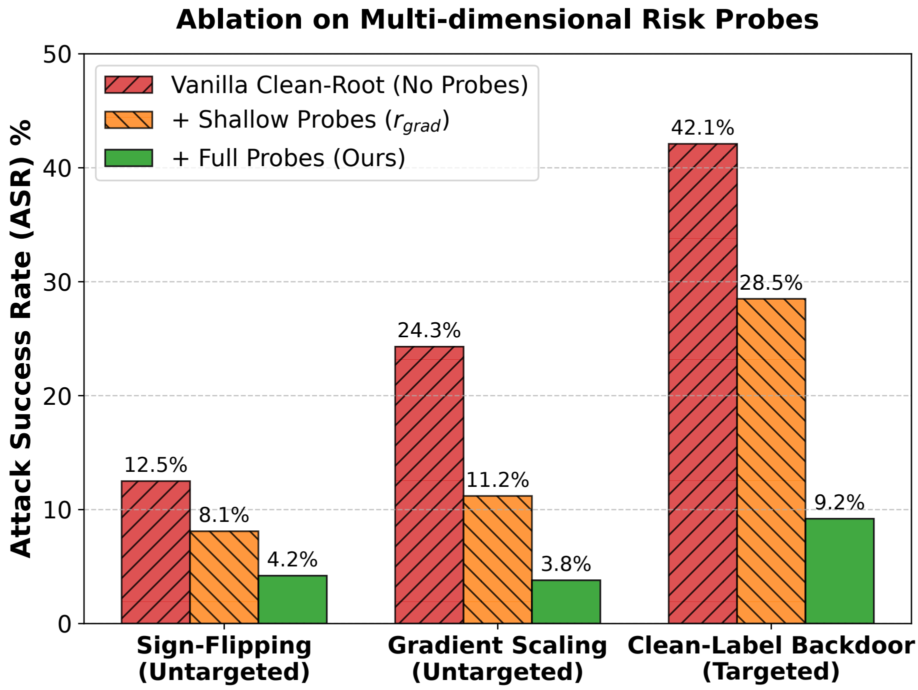

Figure 9: ASR Suppression Comparison of Multi-Dimensional Risk Probes across Attack Types

The visualization clarifies the complementary role of each probe. For Sign-Flipping, whose direction anomaly is explicit, the clean reference already limits ASR to 12.5%, and the shallow gradient probe further reduces it to 8.1%. For Gradient Scaling, where abnormal update magnitude is more salient, the shallow probe reduces ASR from 24.3% to 11.2%. Clean-Label Backdoor is more difficult: its malicious update remains highly collinear with benign directions, so the ASR is 42.1% with only the clean reference and remains 28.5% after adding shallow probes. After enabling deep probes based on cross-entropy (r_{probe}) and high-level feature KL divergence ($r_{trigger}$), the system captures fine-grained functional and representation shifts caused by hidden triggers and reduces the attack success rate to 9.2%.

5.4.3 Layered Gating and Re-Normalization Convergence Benefits

Subsequent to the successful ablation of attacks, securing the rapid restoration of the global model's availability emerges as another pivotal concern. Figure 10 evaluates the disparities in convergence velocity and terminal accuracy among a traditional, indiscriminate global clipping mechanism (Global-Clipping), the execution of layer-wise gating bereft of weight reallocation (Hierarchical w/o Renorm), and the holistic framework introduced herein (Trust-Flow Proposed).

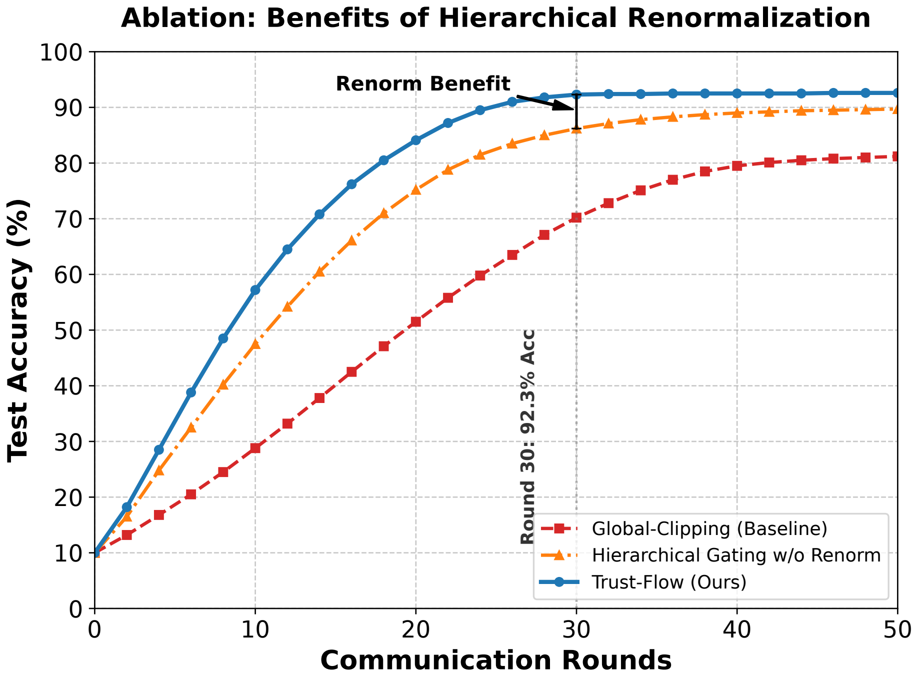

Figure 10: Convergence and Accuracy Benefits of Dynamic Weight Re-Normalization

The trajectory trends starkly indicate that the aggressive global L2 clipping profoundly dilutes the model's generalization capabilities by excessively interfering with the weight updates of the shallow feature extraction layers, effectively stalling the ultimate accuracy at approximately 85%. Although incorporating layer-wise gating ameliorates this deficiency to an extent (elevating accuracy to 89%), the elimination of malicious nodes precipitates a scenario where the effective aggregation weights of the current round fall short of the unit constraint (i.e., $\sum \tilde{w} < 1$). This instigates an "implicit learning rate decay," thereby ensuring the convergence curve remains retarded. The constrained projection and weight re-normalization mechanisms introduced in this study dynamically compensate for the influence of surviving nodes. This strategy enables stable convergence within the 30-round training budget and restores the model accuracy to a level close to the attack-free setting (92.4%), fully preserving system utility while maintaining uncompromising security.

5.5 Sensitivity Analysis on Data Heterogeneity

A plethora of robust aggregation methods lean heavily upon IID conditions. To meticulously evaluate the stability of the proposed framework under varying intensities of heterogeneity, the global accuracy was subjected to testing across four paradigms of Dirichlet distributions ($\alpha = 100, 1.0, \ldots, 0.1$), with the outcomes depicted in Figure 11.

The curves substantiate that within weakly heterogeneous environments dictated by $\alpha = 100$ or $\alpha = 1.0$, Krum and FLTrust exhibit performance rivaling the proposed method (both registering Acc $> 90\%$). However, within extreme long-tail settings ($\alpha \to 0.1$), traditional distance-based methods undergo marked degradation, with accuracy plummeting to a dismal 64.2%. Resiliently,

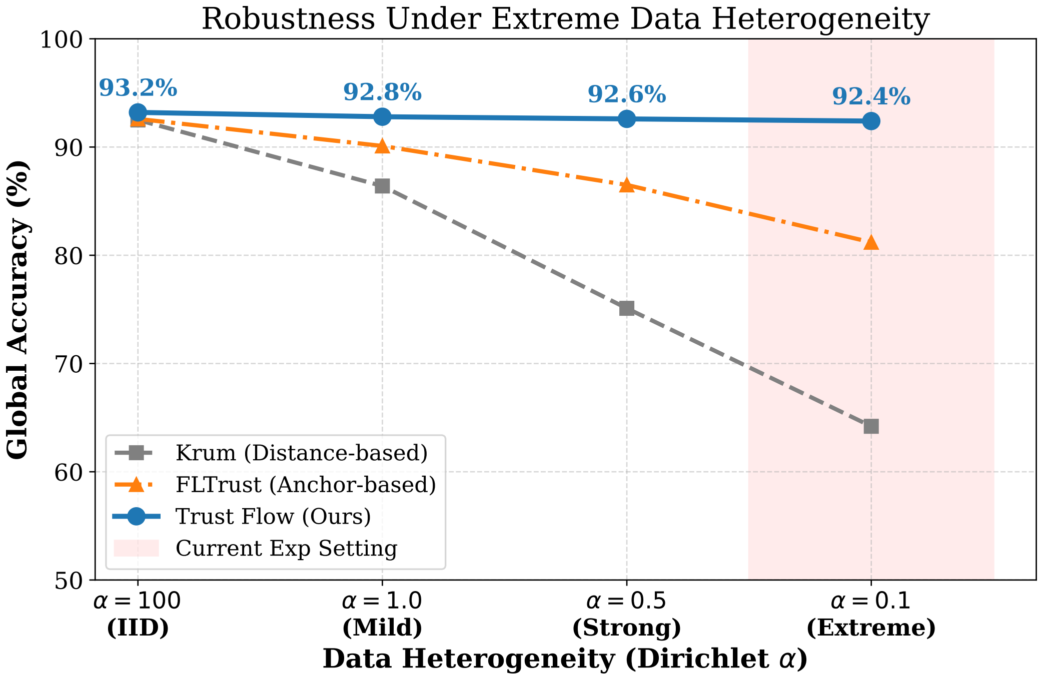

Figure 11: Robustness Stress Test under Varying Data Heterogeneity (Dirichlet α)

the proposed approach stabilizes at approximately 92% under identical adversity, thereby verifying that the dual-stream decoupling architecture possesses superior robustness against profound heterogeneity.

Furthermore, discrete testing at 0.5, 1.0, 2.0, and 3.0 was conducted on the pivotal parameter β_{fusion}. A smaller magnitude fosters the retention of long-tail nodes but simultaneously escalates ASR risks; conversely, an oversized value skews the aggregation excessively towards homogeneous updates. Weighing the trade-offs between Acc and ASR, $\beta_{fusion} = 2.0$ was definitively selected.

5.6 System Computation and Communication Overhead Analysis

Despite harboring a multi-phase evaluation protocol, the computational and communication overhead of the framework remains stringently tractable. Table 6 demonstrates that the edge merely incurs the generation of a TEE Quote and lightweight monitoring, yielding an auxiliary delay of roughly $12 \sim 15$ ms. Concurrently, the communication tier merely appends a TrustReport of approximately 15 KB per transmission. When juxtaposed against the ~ 10 MB scale of model transfer, the bandwidth overhead rests below an imperceptible 0.15%.

Table 6: Edge-Cloud Overhead: Standard FL vs. Proposed Framework (20 Clients)

System Architecture	Edge Extra Latency	Upload Extra Payload	Cloud Aggregation Time
Standard FedAvg	0 ms (Baseline)	0 KB (Baseline)	~ 2.10 s
Trusted FL (Proposed)	**~ 15 ms (TEE signing only)**	**~ 15 KB (appended report)**	**~ 4.85 s (incl. audit & dual-stream)**

Regarding specific costs, the layer-wise cosine evaluation inherently inflates the computational load on the server. In the empirical trials, the aggregation duration per round escalated from 2.10 s to 4.85 s. Given that a single round of communication over a Wide Area Network (WAN) habitually consumes tens of seconds, this nominal computational overhead is entirely acceptable from an engineering perspective.

6 Conclusion and Future Work

Addressing the security vulnerabilities of edge federated learning amidst potent Non-IID constraints and compound attacks, this manuscript formally introduces the Trust-Flow TFL framework, a triad architecture encompassing "hardware-anchored trusted admission, dual-stream reputation evolution, and layer-wise robust aggregation." This framework instantiates a physical root of trust via TEEs and mathematically engineers the synergy of "low false positives paired with high recall" at the algorithmic tier through the orthogonal modeling of $HistPerf$ and $RiskEMA$.

Empirical results dictate that under a 30% mixed attack vector, the framework attains an accuracy of 92.31%(±0.09%) and restricts ASR to 10.21%(±0.05%). Even subjected to the 50% boundary pressure test, it perseveres with steadfast stability, delivering a perfect TPR of 100% alongside an FPR of 0% (metricized by permanent bans). This rigorously confirms the method's robust utility under profoundly adversarial and intensely heterogeneous paradigms.

Limitations and Future Trajectories: Due to the absence of publicly verifiable implementations for several emerging TEE-FL methodologies, this study foregoes their inclusion within the isomorphic comparative baselines. Subsequent phases aim to release the source code and experimental scripts alongside the final manuscript publication, thereby catalyzing community reproduction and expansion. Future endeavors will also endeavor to scale these multidimensional risk probes beyond CV environments into LLM/NLP federated fine-tuning topologies, meticulously validating their generalizability across cross-modal scenarios.

A Detailed Computation of Instant Risk Flow Probes

To augment reproducibility and transparency, this appendix mathematically delineates the calculation details of the core probes ($r_{grad}, r_{probe}, r_{trigger}$) underpinning the instant risk stream originally introduced in Section 4. During the t-th aggregation round, the probes are defined as follows:

A.1 A.1 Gradient Direction and Amplitude Probe (r_{grad})

The r_{grad} probe quantifies the geometric deviation of the client update $\Delta W_k^{(t)}$ relative to the reference direction $g_{root}^{(t)}$, tasked specifically with detecting sign flipping and amplitude anomalies. This metric concurrently evaluates directional deflection (cosine term) and scale mutation (L2 anomaly term):

$$r_{grad,k}^{(t)} = \lambda_d \left(1 - \frac{\Delta W_k^{(t)} \cdot g_{root}^{(t)}}{||\Delta W_k^{(t)}|| \cdot ||g_{root}^{(t)}||}\right) + \lambda_m \max\left(0, \frac{||\Delta W_k^{(t)}||_2 - \mu^{(t)}}{\sigma^{(t)}}\right) \tag{27}$$

Wherein λ_d and λ_m modulate the weighting of the directional and amplitude terms, respectively (defaulting to 0.5 for both); $\mu^{(t)}$ and $\sigma^{(t)}$ signify the mean and standard deviation of the gradient L2 norms across the admitted nodes for round t.

A.2 A.2 Small-Sample Cross-Entropy Probe (r_{probe})

Relying exclusively on distance statistics is grossly inadequate for identifying targeted pollution directed against minority classes. The r_{probe} leverages a small-scale, pristine probe set on the server, denoted as $\mathcal{D}_{probe}$, calculating the incremental surge in cross-entropy loss (Loss Surge) pre- and post-merger:

$$r_{probe,k}^{(t)} = \text{ReLU}\left(\mathcal{L}_{CE}(W_{global}^{(t-1)} + \Delta W_k^{(t)}; \mathcal{D}_{probe}) - \mathcal{L}_{CE}(W_{global}^{(t-1)}; \mathcal{D}_{probe})\right) \tag{28}$$

A.3 A.3 Deep Neuron Backdoor Activation Probe ($r_{trigger}$)

Countering profoundly concealed threats like Clean-Label attacks, standard statistical and validation losses often fall short. The $r_{trigger}$ discerns anomalies through the distributional variance of high-level feature activations. Assuming A_k denotes the client model's pre-activation outputs on the validation set, and $\mathcal{H}(\cdot)$ signifies the Softmax normalization into a probability simplex to satisfy the KL-Divergence prerequisites:

$$r_{trigger,k}^{(t)} = \text{KL-Divergence}\left(\mathcal{H}(A_{base})\|\mathcal{H}(A_k)\right) \tag{29}$$

Post normalization and fusion, these three distinct anomalous signals crystallize into the single-round risk extremum $Risk_k^{(t)}$, rapidly entering the EMA pipeline to empower subsequent fast-circuit breaking adjudication.

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